# Estimation of Mean Squared Errors in the EU SAE Poverty Mapping Package

## Levels and Changes, with and without Benchmarking

October 2026 • revised for 5.2.0-rc.6-wizard.5.14 (w5n)

### Introduction

The bootstrap simulates Gaussian random effects and sampling errors under the fitted model, refits the same estimator, and repeats back-transformation and the final ratio benchmark in reported units. Survey-derived targets are recomputed from the same replicate's Direct estimates in both UFH and MFH. Uploaded targets remain fixed, so their own uncertainty is excluded. Errors are measured against the unbenchmarked synthetic area truth. Auxiliary variables, selected formulas, selected MFH type, populations and estimated sampling covariance inputs are held fixed. Variable selection, MFH3/MFH2 selection, GVF fitting and covariance estimation are not repeated in each draw; their uncertainty is excluded.

The note proceeds from the single-period UFH case (Part I), to the multi-period MFH case and the mean cross-product error (MCPE) needed for changes (Part II), to the change analysis that combines them (Part III). A summary table and practical observations close the note. General treatments of the Fay–Herriot model, EBLUP prediction, REML estimation, MSE, and benchmarking are given in Rao and Molina (2015).

QER revision for 5.2.0-rc.6-wizard.5.14 (w5n), October 2026. This note describes the final-estimator bootstrap and current joint-moment rules. The implementation checks and simulation evidence are documented in docs/guidance/METHODOLOGY_ALIGNMENT.md and docs/validation/MCPE_VALIDATION_STATUS.md. Numerical agreement and Monte Carlo precision do not certify interval coverage for all survey designs. Review model assumptions, boundary sensitivity and any undercoverage before dissemination. Use the latest validation status for current experiment counts and results.

### Part I. The Univariate Fay–Herriot Model (Single Period)

#### *The model and the EBLUP*

Index the small areas (domains) by $d = 1, \ldots, M$, of which $m$ are observed in the survey. The area-level Fay–Herriot model (Fay and Herriot, 1979) combines a sampling layer and a linking layer,

$$\hat{\theta}_d^{dir} = \theta_d + e_d, \theta_d = x_d^{\top} \beta + u_d,$$

The sampling layer has independent normal errors with known working variances; the linking layer has normal area effects and fixed auxiliary covariates. Ordinary UFH variance components are fitted by REML. Log-area-mean UFH uses ML to match emdi bc_sm. The EBLUP combines the Direct and regression predictors with precision-dependent weights.

$$\hat{\theta}_d^{FH} = \hat{\gamma}_d \hat{\theta}_d^{dir} + (1 - \hat{\gamma}_d) x_d^{\top} \hat{\beta}, \hat{\gamma}_d = \frac{\hat{\sigma}_u^2}{\hat{\sigma}_u^2 + \psi_d}.$$

***MSE of the unbenchmarked UFH estimates***

Log mean welfare uses the log of the survey-weighted arithmetic Direct area mean, with sampling variance V/mean². Zero and negative individual welfare values remain in the arithmetic mean; the area mean must be positive and finite. UFH bias correction uses native emdi Slud–Maiti bc_sm with ML fitting and its original-scale analytical MSE. The naive option exponentiates the same fitted log-area mean and uses bootstrap MSE; it does not change the target into a geometric mean. Identity-scale UFH uses the analytical Prasad–Rao/Datta–Lahiri MSE. Arcsine UFH uses the package-compatible point estimator and a consistent original-scale parametric bootstrap.

***Benchmarking and the MSE of benchmarked UFH estimates***

Benchmarking restores aggregate consistency: within each benchmark area $r$ (regions by default), `bench_regional()` applies a ratio adjustment so that the population-weighted mean of the benchmarked estimates reproduces the direct regional aggregate,

$$B_r = \frac{\sum_{d \in r} N_d \hat{\theta}_d^{dir}}{\sum_{d \in r} N_d}, \lambda_r = \frac{B_r}{\sum_{d \in r} \omega_{r,d} \hat{\theta}_d^{FH}}, \hat{\theta}_d^{Bench} = \lambda_r \hat{\theta}_d^{FH} (d \in r),$$

with $\omega_{r,d} = N_d / \sum_{j \in r} N_j$. An external official total can replace $B_r$. A positive ratio factor preserves nonnegativity and ordering only for already nonnegative predictions. It does not repair negative predictions or ensure an upper bound of one; inspect both poverty-rate bounds. An external total is treated as fixed: its own sampling error is not included in the benchmarked MSE, so matching it exactly says nothing about its accuracy.

Consistent with the benchmarking literature (Datta, Ghosh, Steorts, and Maples, 2011; Bell, Datta, and Ghosh, 2013), the uncertainty of the benchmarked predictor is assessed relative to the true small-area value,

$$MSE(\hat{\theta}_d^{Bench}) = E[(\hat{\theta}_d^{Bench} - \theta_d)^2].$$

The package estimates final-estimator MSE by a Gaussian model-based bootstrap. It draws random effects with fitted variance and sampling errors with the fixed working sampling variance, on the model scale. It does not divide empirical residuals by an estimated standard deviation; genuine zero random-effect variance is therefore supported. Each draw has a synthetic area truth and Direct observation:

$$\theta_d^{(b)} = x_d^\top \hat{\beta} + u_d^{(b)},\ y_d^{(b)} = \theta_d^{(b)} + e_d^{(b)},$$

The same estimator is refitted and transformed to the reporting scale. Survey-derived benchmark targets are recomputed from the same replicate Direct values; uploaded targets remain fixed. Benchmarking follows back-transformation, including for log means. Squared errors compare the final original-scale predictor with original-scale latent truth:

$$\widehat{MSE}\left(\hat{\theta}_d^{Bench}\right) = \frac{1}{B}\sum_{b=1}^{B}\left(\hat{\theta}_d^{Bench,(b)} - \theta_d^{(b)}\right)^2.$$

All requested draws are checked for finite predictions and errors. UFH requires at least 90% successful finite replicates, otherwise uncertainty fails explicitly. Successful counts and failure reasons are recorded; MSE = FALSE leaves benchmark MSE unavailable. Final currency benchmarking is repeated inside every replicate, with no post-hoc coefficient-of-variation rescaling.

## Part II. The Multivariate Fay–Herriot Models (Multiple Periods)

***Model variants***

Let $d = 1, \ldots, D$ index domains and $t = 1, \ldots, T$ periods. For each domain, the vector of direct estimates over time is modeled as a regression mean plus correlated random effects plus sampling errors with known $T \times T$ covariance $\Sigma_{e,d}$. Under MFH2 the random effects follow a homoskedastic AR(1) process, $u_{d,t} = \rho\, u_{d,t-1} + a_{d,t}$, so nearby periods share information; MFH1 makes the effects independent across periods; MFH3 allows the AR(1) innovation variance to vary by period. The models are fitted with the `msae` package (Permatasari and Ubaidillah, 2021), following Benavent and Morales (2016).

***Parametric bootstrap for MSE and MCPE (unbenchmarked)***

Selected MFH annual MSEs and cross-year MCPE are computed jointly by parametric bootstrap. Random effects and sampling errors follow the fitted Gaussian covariance model. The same estimator, inverse transformation and optional final benchmark are applied in every accepted draw. MCPE is the expected product of prediction errors; it need not equal their centered covariance.

1. **Random effects.** For MFH2 (homoskedastic AR(1), stationary initialization):

$$u_{d,1}^{(b)} = \frac{a_{d,1}^{(b)}}{\sqrt{1-\hat{\rho}^2}},\ u_{d,t}^{(b)} = \hat{\rho}\, u_{d,t-1}^{(b)} + a_{d,t}^{(b)},\ a_{d,t}^{(b)} \sim N\left(0, \hat{\sigma}_u^2\right).$$

MFH3 uses u0 ~ N(0,1), then u1 = rho × u0 + a1 and the time-varying innovation recursion. MFH1 draws independent effects with a separate variance for each year. These definitions match the fitted covariance, including zero random-effect covariance.

2. **Sampling errors**, drawn jointly across periods, so the temporal covariance of the direct estimates is respected: $e_d^{(b)} \sim N\left(0, \Sigma_{e,d}\right)$.

3. **Synthetic truth and observation:**

$$\theta_{d,t}^{(b)} = x_{d,t}^{\top}\hat{\beta}_t + u_{d,t}^{(b)},\ y_{d,t}^{(b)} = \theta_{d,t}^{(b)} + e_{d,t}^{(b)}.$$

4. Refit the same MFH variant and require usable numerical endpoint checks and finite predictions for all included domains and periods. Failed draws are counted and retried up to the attempt limit. A draw is accepted jointly; all annual MSE and MCPE entries share the same accepted draws and divisor.

With prediction errors $\delta_{d,t}^{(b)} = \hat{\theta}_{d,t}^{(b)} - \theta_{d,t}^{(b)}$,

$$\widehat{MSE}_{d,t} = \frac{1}{n_{valid}}\sum_{b=1}^{n_{valid}} (\delta_{d,t}^{(b)})^2,\ \widehat{MCPE}_d(t_1,t_2) = \frac{1}{n_{valid}}\sum_{b=1}^{n_{valid}} \delta_{d,t_1}^{(b)}\delta_{d,t_2}^{(b)}.$$

The joint second-moment matrix supplies both selected annual MSEs and MCPE. Original point estimates come from the original fit. Requested and successful counts, attempts and Monte Carlo errors are recorded. The shared Bootstrap replicates control defaults to 500 for UFH, MFH and benchmarking, while preserving explicit older saved counts. Select replication after stability checks; a larger count does not guarantee coverage.

The numerical check assesses REML score, local curvature and feasible boundary directions in addition to the package convergence flag. A positive variance alone does not establish convergence. MFH2 is refitted when necessary using the same REML objective, exact gradients and several feasible starts; a usable original fit is preserved and an inferior refit is rejected. A valid zero random-effect covariance is allowed, including in bootstrap refits. The AR(1) correlation is bounded numerically at ±0.999999; an accepted endpoint approximates a limiting stationary fit and warrants sensitivity review. Analytic g3/Fisher-information availability is separately reported. A valid bootstrap can provide uncertainty even when analytic g3 is unavailable.

Boundary-sensitive inference. If the original MFH fit has zero random-effect covariance or reaches the numerical correlation limit, point estimates are retained and inference_status is boundary_sensitive. Normal or delta-method intervals and p-values are provisional: numerical convergence does not establish their coverage. Reports flag this condition and record the count and fraction of accepted bootstrap fits at a boundary. Increasing the bootstrap count reduces Monte Carlo error but does not remedy boundary undercoverage. Review the current validation status and assess model sensitivity before using these results for substantive significance claims.

***Benchmarked MFH estimates***

Benchmarking applies a ratio adjustment separately by group and year to the final reporting-scale estimates, using the requested population coverage and positive aggregation weights:

$$B_{rt} = \frac{\sum_{d\in B_{rt}} N_{d,t}\hat{\theta}_{d,t}^{dir}}{\sum_{d\in B_{rt}} N_{d,t}},\ \lambda_{rt} = \frac{B_{rt}}{\sum_{d\in B_{rt}} \omega_{r,d,t}\hat{\theta}_{d,t}^{EBLUP}},\ \hat{\theta}_{d,t}^{bench} = \lambda_{rt}\hat{\theta}_{d,t}^{EBLUP}.$$

An external regional benchmark can be supplied through `regional_benchmark_mat` instead of being computed from the survey.

With `MSE = TRUE`, the routine runs the same parametric bootstrap as the unbenchmarked case — identical data-generating process, identical refit — with one extra step inside each replicate: the bootstrap EBLUPs are themselves benchmarked before prediction errors are computed,

$$\hat{\theta}_{d,t}^{(b),bench} = \lambda_{rt}^{(b)} \hat{\theta}_{d,t}^{(b)},$$

The survey target in each draw is recomputed from that draw's Direct estimates. Uploaded targets remain fixed. For log means, Direct observations, predictions and truth are first returned to currency units, then benchmarked; MSE and MCPE are computed from final currency-scale errors. These errors are measured against unbenchmarked synthetic truth:

$$\widehat{MSE}_{d,t}^{bench} = \frac{1}{n_{valid}} \sum_{b=1}^{n_{valid}} \left( \hat{\theta}_{d,t}^{(b),bench} - \theta_{d,t}^{(b)} \right)^2, \widehat{MCPE}_{d}^{bench}(t_1, t_2) = \frac{1}{n_{valid}} \sum_{b=1}^{n_{valid}} \delta_{d,t_1}^{(b),bench} \delta_{d,t_2}^{(b),bench}.$$

Benchmarked MSE includes the adjustment's variability and prediction bias under the bootstrap model. The seed and requested replication count are configurable and retained. The same fixed sampling covariance inputs drive the original fit, simulated data and each refit.

## Part III. MSE of Changes in Poverty Rates

A central use of small-area poverty estimates is to test whether poverty has changed between two periods. The error of an estimated change depends not only on the MSE in each period but also on how the errors in the two periods move together; the MCPE captures that covariance. For domain $d$ and periods $t_1, t_2$, the estimated change is

$$\hat{\Delta}_d = \hat{\theta}_{d,t_2} - \hat{\theta}_{d,t_1},$$

and its MSE is computed in `compare_mfh()` (in `02_mfh.R`) as

$$MSE(\hat{\Delta}_d) = MSE_{d,t_1} + MSE_{d,t_2} - 2\,MCPE_d(t_1, t_2).$$

Pointwise 95% intervals and raw p-values remain the primary display. BH and Bonferroni adjusted p-values and flags answer different multiplicity questions. Counts of pointwise results across areas are exploratory and do not have familywise error control. Positive MCPE reduces the change MSE when annual MSEs are held fixed, but its sign is estimated, not assumed. Unavailable or invalid joint moments produce unavailable inference.

The same helper serves both the unbenchmarked change analysis (fed by the MSE/MCPE of Part II's unbenchmarked bootstrap) and the benchmarked one (fed by mse_bench / mcpe_bench), so the change formula, unadjusted confidence intervals, raw, BH-adjusted and Bonferroni-adjusted p-values, and the corresponding significance fields are computed consistently in the two cases.

Joint-moment guard. All three moments must come from the same accepted draws. A valid second-moment matrix obeys |MCPE| ≤ sqrt(MSE1 × MSE2), up to roundoff. The implementation checks this bound and finite nonnegative diagonal entries. It does not replace invalid moments with independence.

MFH: invalid joint moments ⇒ unavailable uncertainty; Direct/UFH: MCPE = 0 is an explicit period-independence assumption.

Unavailable moments, invalid joint moments and zero inferential variance are recorded with diagnostic status and unavailable intervals or p-values. A higher bootstrap count may reduce simulation error, but does not repair model misspecification. Period independence for Direct and UFH requires separate justification when surveys overlap.

MFH log models also transform the arithmetic Direct area means. Corrected currency predictions use the Gaussian conditional mean exp(eta + v/2), where v is the diagonal of $G - G(G + R)^{-1}G$. The naive option uses exp(eta). This multivariate extension is distinct from emdi's univariate Slud–Maiti method; saved MFH settings named bc_sm are accepted as a legacy alias. Original-scale MSE and MCPE are estimated after inverse transformation inside the bootstrap. Currency changes use the joint currency moments directly:

$$MSE(\Delta) = MSE_1 + MSE_2 - 2\ MCPE_{12}$$

Percentage mean-welfare change is 100 × (mean2/mean1 − 1). Its log-ratio variance is approximately $MSE1/mean1^2 + MSE2/mean2^2 - 2\ MCPE/(mean1 \times mean2)$, using the same original-scale joint bootstrap moments for MFH and explicit MCPE = 0 for UFH. The interval is formed on the log-ratio scale and exponentiated. Missing legacy MFH moments require a rerun; they are never reconstructed by mixing analytical level MSE with bootstrap change MSE. Currency differences remain in fields ending in _eur.

## Summary of Methods

| Estimate | Unbenchmarked MSE | Benchmarked MSE |
|---|---|---|
| UFH level identity / corrected log | Identity: analytical Prasad–Rao/Datta–Lahiri; log bc_sm: native emdi Slud–Maiti (ML) | Gaussian full-estimator bootstrap; final benchmark in reporting units each draw |
| UFH level arcsine / naive log | Original-scale bootstrap matching the point estimator | Same bootstrap with final original-scale benchmark |
| MFH level (MFH1/2/3) | Joint parametric bootstrap; shared Bootstrap replicates control, default 500 | Same final-estimator bootstrap with reporting-scale benchmark each draw |
| MFH change $t_1 \rightarrow t_2$ | $MSE_{t_1} + MSE_{t_2} - 2\,MCPE$ via `compare_mfh()` | Same coherent joint formula and validity check for benchmarked moments |

The bootstrap simulates Gaussian random effects and sampling errors under the fitted model, refits the same estimator, and repeats back-transformation and the final ratio benchmark in reported units. Survey-derived targets are recomputed from the same replicate's Direct estimates in both UFH and MFH. Uploaded targets remain fixed, so their own uncertainty is excluded. Errors are measured against the unbenchmarked synthetic area truth. Auxiliary variables, selected formulas, selected MFH type,

populations and estimated sampling covariance inputs are held fixed. Variable selection, MFH3/MFH2 selection, GVF fitting and covariance estimation are not repeated in each draw; their uncertainty is excluded.

## Practical Notes and Observations

Benchmarking redistributes information; it does not create it. Its value rests on the benchmark target being more reliable than the individual model estimates, which holds when the regional direct aggregate pools enough sampled units to be precise. Where a region–period cell contains only a few sampled domains, the direct benchmark is itself noisy and the ratio adjustment transmits that noise to every domain in the cell; a coarser benchmark geography or an external official total is then preferable. Because the adjustment factors $\lambda_{rt}$ are computed independently across periods, volatile direct regional aggregates can also inject artificial volatility into the estimated trend even when the model estimates are stable.

Bootstrap accuracy depends on the fitted model and fixed sampling covariance approximating the data-generating process. Failed refits and insufficient successful draws require review. The implementation checks and simulation evidence are documented in docs/guidance/METHODOLOGY_ALIGNMENT.md and docs/validation/MCPE_VALIDATION_STATUS.md. Numerical agreement and Monte Carlo precision do not certify interval coverage for all survey designs. Review model assumptions, boundary sensitivity and any undercoverage before dissemination. Use the latest validation status for current experiment counts and results.

Comparing benchmarked and unbenchmarked uncertainty

1. Both calculations use the shared full-estimator bootstrap machinery. Record seeds and successful-draw sets when comparing MSEs: failed refits or different requested counts can change the accepted draws. Differences should be assessed with the exported Monte Carlo diagnostics.

Comparable-over-time small-area estimation is an active area of methodological work (Molina and Romero, 2025).

## References


Bell, W. R., Datta, G. S., and Ghosh, M. (2013). Benchmarking small area estimators. *Biometrika*, 100(1), 189–202.

Benavent, R. and Morales, D. (2016). Multivariate Fay–Herriot models for small area estimation. *Computational Statistics & Data Analysis*, 94, 372–390.

Datta, G. S., Ghosh, M., Steorts, R., and Maples, J. (2011). Bayesian benchmarking with applications to small area estimation. *Test*, 20(3), 574–588.

Datta, G. S. and Lahiri, P. (2000). A unified measure of uncertainty of estimated best linear unbiased predictors in small area estimation problems. *Statistica Sinica*, 10(2), 613–627.

Slud, E. V. and Maiti, T. (2006). Mean-squared error estimation in transformed Fay–Herriot models. Journal of the Royal Statistical Society, Series B, 68(2), 239–257.

Fay, R. E. and Herriot, R. A. (1979). Estimates of income for small places: an application of James–Stein procedures to census data. *Journal of the American Statistical Association*, 74(366), 269–277.

Harmening, S., Kreutzmann, A.-K., Schmidt, S., Salvati, N., and Schmid, T. (2023). A Framework for Producing Small Area Estimates Based on Area-Level Models in R. *The R Journal*, 15(1), 316–341.

Kreutzmann, A.-K., Pannier, S., Rojas-Perilla, N., Schmid, T., Templ, M., and Tzavidis, N. (2019). The R package emdi for estimating and mapping regionally disaggregated indicators. *Journal of Statistical Software*, 91(7), 1–33.

Molina, I. and Romero, E. (2025). Comparable Small Area Estimates over Time. Working document.

Permatasari, N. and Ubaidillah, A. (2021). msae: an R package of multivariate Fay–Herriot models for small area estimation. *The R Journal*, 13(2), 111–122.

Prasad, N. G. N. and Rao, J. N. K. (1990). The estimation of the mean squared error of small-area estimators. *Journal of the American Statistical Association*, 85(409), 163–171.

Rao, J. N. K. and Molina, I. (2015). *Small Area Estimation*, 2nd edition. Wiley, Hoboken.